Supplementary Information

Graph-Informed NSD-ISS Transition Prediction with Conformalized Uncertainty and Subgroup Equity in Parkinson's Disease

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S-1. Reserved

Reserved for the four-method conformal-ablation table (IPCW / marginal / naive / Bonferroni) at 90% and 95% CL. Numerical content is reported in the main-text Methods (§Cause-specific conformal CIF bands with IPCW) and the Results figures.

S-2. Reserved

Reserved for the competitor-comparison table positioning this work against Candès *et al.* 2023 and Sreenivasan *et al.* 2025. Discussion is in the main-text Discussion (§Positioning relative to prior work).

S-3. Carrier Subgroup Analysis

Background and pre-registration

The original Paper 3+4 submission excluded LRRK2 and GBA carrier subgroups from inferential analysis under the rationale that per-fold $n < 10$, the result of a silent feature-extraction bug in `scripts/assemble_paper1_features.py:extract_genetics()` that failed to match the IU Genetic Consensus variant-name encoding (`G2019S`, `R1441G`, `N409S`, `L483P`). After the bug fix in commit `672b439`, the post-fix carrier columns in `features.paper1_features_with_targets` are: 175 LRRK2 carriers, 111 GBA carriers, and 441 APOE $\varepsilon4$ carriers. Strata are defined as mutually exclusive in this analysis: LRRK2+ includes any patient with `lrrk2_carrier=1` (including dual carriers); APOE $\varepsilon4$+ only and GBA+ only require the other two carrier flags to be 0. The pre-registration locking the hypotheses (H1/H2/H3), MIN_SUBGROUP_SIZE_CARRIER =

50, and decision rule was committed on 2026-04-23 *before* any compute on the bug-fixed feature table (`outputs/paper4/subgroup_carriers/PRE_REGISTRATION.md`). The runner script is `scripts/paper3plus4/run_subgroup_with_lrrk2_gba_fix.py`.

Cohort sizes (post mutual-exclusivity definition)

Stratum-level (cohort-wide) sizes versus 5-fold cross-validation per-fold mean and range:

Table S1: **Carrier-stratum cohort and per-fold sizes.** Cohort-level n is the post-bug-fix carrier flag count in `features.paper1_features_with_targets` after applying mutual-exclusivity definitions. Per-fold n is the count of test patients within that stratum across the 5 stratified cross-validation folds (mean and min–max range). Per-fold events count distinct first-event records used in the time-dependent concordance computation.

Stratum	Cohort n	Per-fold n (mean)	Per-fold n (range)	Per-fold events (m
LRRK2+ (incl. dual carriers)	175	140.2	109–179	
APOE ε4+ only	375	161.0	109–175	
GBA+ only	79	50.0	10–92	
Non-carrier (reference)	1,547	590.4	567–636	

H1 — Per-stratum C-td (fairness on discrimination)

Per-stratum time-dependent concordance averaged across 5 stratified cross-validation folds, with 95% patient-level bootstrap CIs (1,000 resamples). The CI envelope columns report the minimum lower bound and maximum upper bound across the 5 per-fold CIs (a conservative summary across folds rather than a refit-bootstrap). All carrier strata's CI envelopes overlap the non-carrier reference envelope, supporting the H1 fairness pre-registration.

Table S2: **H1 — Per-stratum C-td (mean ± std across 5 folds; CI envelope across 5 per-fold patient-bootstrap 95% CIs).** Per-fold n and event counts vary by fold composition; the headline n in column 5 is the per-fold mean. The GBA+ only entry has elevated variance because per-fold n ranges from 10 to 92; the wide CI envelope [0.500, 1.000] reflects the smallest per-fold sample size.

Model	Stratum	C-td (mean ± std)	CI envelope (lo, hi)	n (per-fold mean)	Events
DeepHit	LRRK2+	0.911 ± 0.024	[0.818, 0.976]	140.2	
	APOE ε4+ only	0.932 ± 0.026	[0.811, 1.000]	161.0	
	GBA+ only	0.862 ± 0.183	[0.500, 1.000]	50.0	
	Non-carrier	0.928 ± 0.016	[0.867, 0.956]	590.4	
Graph-DT	LRRK2+	0.897 ± 0.026	[0.778, 0.954]	140.2	
	APOE ε4+ only	0.907 ± 0.043	[0.776, 0.990]	161.0	
	GBA+ only	0.854 ± 0.184	[0.500, 1.000]	50.0	
	Non-carrier	0.906 ± 0.033	[0.806, 0.954]	590.4	

Verdict. H1 PASS: every carrier-stratum CI envelope overlaps the non-carrier reference for both DeepHit and Graph-DT.

H2 — Bootstrap interaction tests (BH-FDR-corrected)

Reviewer #3 (reviewer3.com) flagged that combining per-fold interaction p-values via Fisher's $\chi^2_{2k} = -2\sum \ln(p_i)$ violates the procedure's strict independence assumption: in a 5-fold CV scheme the test sets are disjoint but the trained models share 60% of the training data, so per-fold p-values are positively correlated and Fisher's combination yields spuriously low pooled p-values. The fix (WS-P3-CRIT-B) replaces the per-fold-then-Fisher-combine pattern with a SINGLE patient-level bootstrap interaction test (B=500, `random_state=42`) on the pooled out-of-fold predictions across the 5 folds: each patient appears in exactly one test fold by construction, so a single permutation test on the pooled cohort is the correct statistic for $H_0 : \Delta C_{\text{td}} = 0$. The pooled-OOF inference framework follows Vovk *et al.* [3]. BH–FDR correction is applied across the 6-cell carrier grid (3 strata × 2 models) using `scipy.stats.false_discovery_control(method='bh')`. The legacy per-fold-then-Fisher pipeline is retained in the JSON record under `p_fisher_legacy` and `p_fdr_fisher_legacy` for the side-by-side audit shown below; pre-fix outputs are snapshotted at `outputs/paper4/subgroup_carriers/_pre_crit_b/`.

Table S3: **H2 — Bootstrap interaction tests with BH-FDR correction (post-WS-P3-CRIT-B).** $\Delta C_{\text{td}}^{\text{pooled}}$ is computed on the union of the 5 per-fold OOF predictions (carrier C-td − non-carrier C-td); negative values mean the carrier stratum performs worse than the non-carrier reference. p_{pooled} is the new pooled-OOF permutation p-value (CRIT-B canonical); p_{Fisher} is the legacy Fisher-combined value (no longer drives FDR); $p_{\text{FDR,pooled}}$ and $p_{\text{FDR,Fisher}}$ are the BH-corrected values from each respective pipeline. Numbers in this table are pulled directly from `outputs/paper4/subgroup_carriers/interaction_tests_carriers.json` (post-fix) and `outputs/paper4/subgroup_carriers/_pre_crit_b/interaction_tests_carriers.json` (pre-fix).

Model	**Stratum**	$\Delta C_{\text{td}}^{\textbf{pooled}}$	p_{pooled} **(CRIT-B)**	p_{Fisher} **(legacy)**	$p_{\text{FDR,pooled}}$	$p_{\text{FDR,Fisher}}$
DeepHit	LRRK2+	−0.017	0.120	0.236	0.719	0.372
DeepHit	GBA+ only	+0.004	0.856	0.355	0.856	0.426
DeepHit	APOE ε4+ only	+0.003	0.774	0.683	0.856	0.683
Graph-DT	LRRK2+	−0.014	0.273	0.163	0.737	0.372
Graph-DT	GBA+ only	−0.017	0.431	0.049	0.737	0.295
Graph-DT	APOE ε4+ only	−0.009	0.491	0.248	0.737	0.372

Verdict. The H2 PASS verdict is preserved under the methodologically-correct pooled-OOF re-test. All six $p_{\text{FDR,pooled}}$ values exceed the pre-registered α=0.05 threshold, with the smallest equal to 0.719 (DeepHit×LRRK2+, $\Delta C_{\text{td}}=-0.017$). The legacy Fisher pipeline produced a marginal Graph-DT×GBA+ only p_{Fisher}=0.049 that did not survive FDR correction; the pooled-OOF re-test relabels this borderline cell as p_{pooled}=0.431 ($p_{\text{FDR,pooled}}$=0.737), which is the correct inferential statistic for dependent per-fold p-values and confirms the pre-fix p_{Fisher}=0.049 was a multiple-comparison artifact amplified by the positive dependence between CV folds (the failure mode reviewer3.com #3 anticipated). The §S-CRIT-B audit table below reports the per-cell change.

H3 — Conditional conformal coverage (per stratum)

Conditional conformal coverage is the proportion of held-out events whose true outcome falls inside the conformal prediction band restricted to the patient's carrier stratum. Pre-registration: stratum-conditional coverage at the 90% confidence level should fall within ± 0.03 of the empirical marginal coverage. **The empirical marginal coverage shifted from** 0.82 **(pre-fix) to** 0.90 **(post-fix) after the WS-P3-CRIT-A IPCW formula correction (commit `f8a7c54`, see § S-CRIT-A below): per-marginal-cohort coverage rose from** 0.817 **to** 0.902 **for DeepHit and from** 0.818 **to** 0.905 **for Graph-DT at 90% CL.** Under the corrected IPCW pipeline, the H3 tolerance window is therefore $[0.87, 0.93]$ at the 90% CL.

Table S4: **H3 — Conditional conformal coverage per stratum (post-WS-P3-CRIT-A IPCW formula fix).** Mean across 5 folds at the 90% and 95% confidence levels. The pre-registered H3 tolerance is ± 0.03 around the empirical marginal coverage; under the corrected IPCW pipeline (commit `f8a7c54`) the empirical marginal at 90% CL is $\approx$ 0.90, so the per-stratum tolerance window is $[0.87, 0.93]$ at 90% CL and $[0.92, 0.98]$ at 95% CL. **All eight (model×stratum) cells fall within tolerance at both confidence levels.**

Model	Stratum	Mean coverage (90% CL)	Mean coverage (95% CL)
DeepHit	LRRK2+	0.898	0.949
	GBA+ only	0.919	0.960
	APOE ε4+ only	0.909	0.953
	Non-carrier	0.901	0.950
Graph-DT	LRRK2+	0.894	0.951
	GBA+ only	0.926	0.962
	APOE ε4+ only	0.915	0.955
	Non-carrier	0.906	0.953

Verdict. H3 PASS: stratum-conditional coverage at the 90% CL lies within $[0.87, 0.93]$ for all (model×stratum) cells. The largest single deviation from nominal 0.90 is Graph-DT×GBA+ only at +0.026 (still inside the ± 0.03 window); the smallest is DeepHit×Non-carrier at +0.001. The pre-fix H3 FAIL verdict (mean stratum coverage 0.795–0.835) was an artifact of the silent IPCW formula bug (per-fold weight $1/G(C_i)$ for censored survivors instead of the Candès–Lei–Ren three-case rule), not of the carrier strata themselves — the same bug that depressed pooled marginal coverage by $\approx$ 9 pp at 90% CL also depressed every stratum's coverage by a similar amount. Under the corrected IPCW pipeline the carrier strata behave as well as the demographic (sex/age) strata.

Mondrian per-stratum sensitivity check (WS-P3-14b)

To verify that the H3 PASS is not merely a coincidence of the post-CRIT-A marginal lift, we performed a Mondrian per-stratum recalibration [1, 3] as a sensitivity check: for each fold and each carrier stratum, the IPCW conformal quantiles were re-estimated on the calibration set restricted to that stratum, then applied to the eval-set patients of the same stratum. This is the statistically correct deployment-time procedure for genetically-defined subpopulations. The runner is at `scripts/paper3plus4/run_mondrian_carriers.py`; with `MIN_CAL_PER_STRATUM=30` the GBA+ only stratum is Mondrian-eligible in only 2 of 5 folds (per-fold cal-N $\in \{43, 49, 87, 87, 98\}$ across folds; the remaining folds fall back to the marginal quantile). LRRK2+ and APOE ε4+ only are eligible in all 5 folds.

Table S5: **Mondrian sensitivity check (90% CL, post-CRIT-A IPCW pipeline).** Marginal column reproduces Table S4; Mondrian column applies stratum-specific quantiles. Δ is Mondrian $-$ marginal. Width inflation reports the relative increase in mean band width under Mondrian. The GBA+ only stratum is Mondrian-eligible in 2/5 folds only (`n_cal` < 30 in the other folds); other strata 5/5.

Model	Stratum	Marginal	Mondrian	Δ	Width inflation	Folds (Mondrian)
DeepHit	LRRK2+	0.898	0.907	+0.009	+141%	5/5
	GBA+ only	0.919	0.938	+0.037	+259%	2/5
	APOE ε4+ only	0.909	0.912	+0.003	+41%	5/5
	Non-carrier	0.901	0.898	−0.003	+7%	5/5
Graph-DT	LRRK2+	0.894	0.915	+0.020	+72%	5/5
	GBA+ only	0.926	0.933	+0.028	+161%	2/5
	APOE ε4+ only	0.915	0.919	+0.004	+27%	5/5
	Non-carrier	0.906	0.901	−0.005	+5%	5/5

Sensitivity-check verdict. Mondrian recalibration confirms the H3 PASS: all 8 (model$\times$stratum) cells stay within the $[0.87, 0.93]$ tolerance window (largest Mondrian deviation +0.038 for DeepHit$\times$GBA+ only). Mondrian provides a small per-stratum coverage boost ($\Delta \in [-0.005, +0.037]$) but at substantial width cost (mean band width inflates +5% for Non-carrier to +259% for the data-poor DeepHit$\times$GBA+ only cell, where the per-fold Mondrian quantile is computed from only $\approx$ 45 calibration patients). The observed inflation pattern is the expected statistical-efficiency cost of Mondrian (Boström *et al.* 2025): smaller calibration N at fixed nominal coverage produces wider bands. The GBA+ only stratum's 2/5 fold eligibility is the fundamental finite-sample limit and is the cleanest motivation for multi-site pooling (PPMI + PDBP + UK Biobank-PD) for clinical deployment in the GBA+ subpopulation. The full per-(fold, model, CL, stratum) Mondrian record is at `outputs/paper4/subgroup_carriers/conditional_coverage_carriers_mondrian.json` and the aggregate summary at `outputs/paper4/subgroup_carriers/mondrian_summary.json`.

Honest disclosure and deployment recommendation

The pre-registered overall verdict for WS-P3-14 (post-CRIT-A + post-CRIT-B + post-WS-P3-14b) is therefore **full pass**: H1 fairness is supported (per-stratum C-td CIs overlap the non-carrier reference), H2 is supported (post-CRIT-B pooled-OOF interaction tests yield no significant model$\times$subgroup interaction after BH-FDR correction; smallest p_{FDR}=0.719), and H3 is supported (post-CRIT-A IPCW marginal coverage within ±0.03 of nominal across all 8 strata cells, and the Mondrian sensitivity check confirms the same conclusion at significantly inflated band width). For *deployment* to genetically-defined subpopulations we recommend Mondrian recalibration in any setting where the per-stratum nominal coverage guarantee outweighs the band-width cost; the marginal calibration is sufficient for population-level reporting. A multi-site pooled analysis would lift the GBA+ only stratum out of the per-fold $n_{\text{cal}} < 30$ regime, allowing 5/5 fold Mondrian eligibility, and is the natural next step for this work.

Source artefacts

- `outputs/paper4/subgroup_carriers/PRE_REGISTRATION.md` — locked hypothesis and decision rule (2026-04-23).
- `outputs/paper4/subgroup_carriers/subgroup_ctd_carriers.json` — per-fold per-stratum C-td with bootstrap CIs.
- `outputs/paper4/subgroup_carriers/interaction_tests_carriers.json` — per-fold bootstrap interaction tests + Fisher combination.
- `outputs/paper4/subgroup_carriers/conditional_coverage_carriers.json` — per-fold stratum-conditional coverage at 90% and 95% CL.
- `outputs/paper4/subgroup_carriers/decision_verdict.json` — pooled summary, FDR-corrected p-values, H3 deviation list.
- `outputs/paper4/subgroup_carriers/CLAIMS.md` — claim-mapping audit record.
- `scripts/paper3plus4/run_subgroup_with_lrrk2_gba_fix.py` — runner.

S-CRIT-A. IPCW Formula Correction (Reviewer Round 2 #2)

Statement of the bug and correct formula

Reviewer #2 (npj-DM, Round 2) flagged that the original implementation of the cause-specific IPCW conformal weights violated Candès, Lei, and Ren [2]. Specifically, the per-observation weight at evaluation horizon t_j was computed as $w_i = 1/\hat{G}(C_i)$ for censored survivors with $C_i \geq t_j$ and $w_i = 1$ for uncensored events with $T_i \leq t_j$. Both choices are incorrect.

The Candès 2023 formula for fixed-time conformal evaluation is, for each calibration observation $i = (X_i, \delta_i)$ contributing to horizon t_j:

$$w_i^{(k,j)} = \begin{cases} 1/\hat{G}(T_i^-) & \delta_i = 1,\ T_i \leq t_j \quad \text{(uncensored event before } t_j\text{)} \\ 1/\hat{G}(t_j^-) & X_i > t_j \quad \text{(at risk past } t_j\text{; censored or uncensored)} \\ \text{excluded} & \delta_i = 0,\ C_i < t_j \quad \text{(censored before } t_j\text{)} \end{cases} \tag{1}$$

where $X_i = \min(T_i, C_i)$, $\delta_i \in \{0,1\}$, T_i is the event time, C_i the censoring time, and $\hat{G}$ the Kaplan–Meier estimate of the censoring survival function. The original (buggy) implementation used the patient's own censoring time C_i for survivors, which over-weights patients censored late, and unit weight for uncensored events, which under-weights events whose at-risk-process realisation is short.

The corrected helper `compute_per_observation_ipcw_weight()` in `src/giman_pipeline/paper4/conformal` implements Eq. 1. The same fix was mirrored in `src/giman_pipeline/paper4/calibration.py`. The KM estimation, the score formula, the per-(cause, time) quantile rule, and the random-state are unchanged; the only modification is the per-observation weight.

Pre/post-correction marginal coverage (5-fold mean ± SD)

Table S6: **Pre/post-correction marginal conformal coverage** on all 10 Paper 3 checkpoints (5 DeepHit + 5 Graph-DT), 50/50 calibration/evaluation split per fold, identical seeds. With the corrected IPCW formula, both models meet the nominal target at the 80%, 90%, and 95% confidence levels, closing the previously-documented structural undercoverage at 90% CL.

Model	CL	Pre coverage	Post coverage	Δ (pp)
DeepHit	80%	0.627 ± 0.048	0.797 ± 0.010	+16.97
	90%	0.817 ± 0.025	0.902 ± 0.008	+8.47
	95%	0.911 ± 0.015	0.951 ± 0.006	+3.93
Graph-DT	80%	0.626 ± 0.040	0.798 ± 0.009	+17.13
	90%	0.818 ± 0.024	0.905 ± 0.007	+8.67
	95%	0.914 ± 0.013	0.953 ± 0.006	+3.88

Per-cause coverage shifts at 90% CL

Table S7: **Per-cause marginal coverage at 90% CL, pre vs. post correction.** 5-fold mean. After correction every cause-specific value lies within ±0.01 of the nominal 0.90 target for both models. Cause indices 0, 1, 2, 3, 4, 5, 6 correspond to NSD-ISS stages 0, 1, 2B, 3, 4, 5, 6 respectively (the destination of the next observed transition).

Cause	DeepHit pre	DeepHit post	Graph-DT pre	Graph-DT post
0 (Stage 0)	0.825	0.902	0.823	0.904
1 (Stage 1)	0.820	0.891	0.826	0.906
2 (Stage 2B)	0.820	0.903	0.816	0.902
3 (Stage 3)	0.816	0.901	0.810	0.900
4 (Stage 4)	0.807	0.910	0.813	0.912
5 (Stage 5)	0.815	0.908	0.819	0.912
6 (Stage 6)	0.818	0.897	0.824	0.901

Magnitude of change discussion

The correction is empirically substantial: marginal coverage at the 90% confidence level shifts by +8.5 pp (DeepHit) and +8.7 pp (Graph-DT), elevating both models from ~ 0.82 (the previously-reported "structural undercoverage") to the nominal ~ 0.90 target. At 95% CL the shift is +3.9 pp for both models; at 80% CL it is +17 pp. These magnitudes exceed the 0.005 reporting threshold pre-registered in the WS-P3-CRIT-A workstream, so the abstract, the main-text conformal-coverage table, and Methods §Cause-specific conformal CIF bands with IPCW have all been updated in the main manuscript.

The previously-reported "structural concentration of CIF values near zero" framing for the 90%-CL undercoverage is now obsolete: the gap to nominal was an implementation bug, not an inherent property of pointwise conformal bands on competing-risks CIFs. The forward-vs-backward directional gap also narrows from 7.0 pp (pre) to 5.7 pp (post), because forward coverage is now lifted closer to nominal while backward coverage—which is intrinsically harder due to treatment-driven censoring informativeness—remains below 0.90 (0.854 mean across folds). The substantive Discussion claim that backward-transition coverage is sub-nominal is preserved; only the numerical values were updated.

The mean band widths grew with the corrected formula: at 90% CL, DeepHit width $0.0047 \to 0.0191$ and Graph-DT width $0.0171 \to 0.0478$. This is the price of correctness: the original buggy weights produced narrower bands by under-weighting. The post-correction bands are still substantially narrower than the Bonferroni baseline (~ 0.765, ~ 16–$40\times$ wider) and remain clinically interpretable given that the units are CIF-probability (so widths ~ 0.05 correspond to ± 5 percentage points of cumulative incidence).

Files modified

- `src/giman_pipeline/paper4/conformal_survival.py`: added `compute_per_observation_ipcw_weight(` implementing Eq. 1; modified `CauseSpecificConformal.calibrate()` to call it for each (i, t_j) pair.
- `src/giman_pipeline/paper4/calibration.py`: replaced the same buggy weight pattern in `_compute_observed_status()` with the corrected formula; affects ECE, reliability diagrams, and Hosmer–Lemeshow test inputs.
- `tests/paper4/test_conformal_ipcw.py` (new): three TDD unit tests on a synthetic Kaplan–Meier with hand-computed $G(t)$ values, plus a smoke-test on a small synthetic CIF problem. All four tests fail on the pre-fix code (they cannot import `compute_per_observation_ipcw_weight`) and pass after the fix.
- Pre-fix conformal results snapshotted to `outputs/paper4/conformal/_pre_ipcw_fix/` for the pre/post comparison above.

S-CRIT-B. Pooled-OOF Single Test Replaces Fisher's Combined p-values (Reviewer Round 2 #3)

Statement of the bug

Reviewer #3 (npj-DM, Round 2) observed:

> "Combining p-values across cross-validation folds via Fisher's method violates the procedure's strict independence assumption. ... Fisher's method (using the test statistic $-2\sum \ln(p_i) \sim \chi^2_{2k}$) requires the individual p-values to be independent. In a 5-fold cross-validation scheme, while the test sets are disjoint, the models evaluated on these test sets are trained on highly overlapping training data (sharing 60% of the total dataset). ... A method designed for dependent p-values or a single test on the pooled out-of-fold predictions is required."

The pre-fix runner (`scripts/paper3plus4/run_subgroup_with_lrrk2_gba_fix.py`, lines 603–624) aggregated 5 per-fold permutation p-values per (model, carrier-stratum) cell and applied Fisher's combination

$$\chi^2_{2k} = -2\sum_{i=1}^{k} \ln(p_i), \qquad p_{\text{Fisher}} = P(\chi^2_{2k} \geq \chi^2_{\text{obs}}), \tag{2}$$

followed by BH–FDR correction across the resulting 6 cells. Because the per-fold predictions arose from CV models sharing 60% of their training data, the per-fold p_i are positively correlated, and Eq. 2 systematically yields spuriously low pooled p-values relative to the true joint distribution.

Correct procedure (pooled-OOF single test)

For each (model, carrier-stratum) pair the corrected procedure applies a single patient-level bootstrap interaction test on the union of the 5 per-fold OOF predictions. Specifically, with $\mathcal{P}_k$ denoting the disjoint test patnos in fold k and $A_k(\cdot)$ the carrier vs reference assignment within that fold:

$$\Delta C_{\text{td}}^{\text{pooled}} \;=\; C_{\text{td}}\left(\bigcup_{k=1}^{5}\{i \in \mathcal{P}_k : A_k(i) = \text{carrier}\}\right) - C_{\text{td}}\left(\bigcup_{k=1}^{5}\{i \in \mathcal{P}_k : A_k(i) = \text{Non-carrier}\}\right), \tag{3}$$

with the two-sided permutation p-value

$$p_{\text{pooled}} \;=\; \frac{1 + \sum_{b=1}^{B} \mathbb{1}\Big[\,|\Delta^*_b| \geq |\Delta C_{\text{td}}^{\text{pooled}}|\,\Big]}{1 + B}, \tag{4}$$

where Δ^*_b is the bootstrapped statistic obtained by permuting the (carrier vs reference) labels within the pooled cohort. We use B=500 permutations and `random_state=42`. Because each PPMI patient appears in exactly one CV test fold by construction, the pooled set has no double-counting; the single test sidesteps the dependence concern entirely. The general theoretical framework for such pooled-OOF inference is given in Vovk *et al.* [3] (cross-conformal prediction, Chapter 4). BH–FDR correction is then applied across the same 6-cell grid.

Pre/post comparison per (model, stratum) cell

Table S8: **Pre/post WS-P3-CRIT-B comparison.** The pre-fix Fisher-combined p-values are taken from `outputs/paper4/subgroup_carriers/_pre_crit_b/interaction_tests_carriers.json`; the post-fix pooled-OOF p-values from `outputs/paper4/subgroup_carriers/interaction_tests_carriers.jso` Per-fold predictions, fold splits, and `random_state=42` are unchanged across the two pipelines; only the cross-fold combination method differs.

Model	**Stratum**	p_{Fisher} **(pre)**	p_{pooled} **(post)**	$p_{\text{FDR,Fisher}}$ **(pre)**	$p_{\text{FDR,pooled}}$ **(post)**
DeepHit	LRRK2+	0.236	0.120	0.372	0.719
DeepHit	GBA+ only	0.355	0.856	0.426	0.856
DeepHit	APOE ε4+ only	0.683	0.774	0.683	0.856
Graph-DT	LRRK2+	0.163	0.273	0.372	0.737
Graph-DT	GBA+ only	0.049	0.431	0.295	0.737
Graph-DT	APOE ε4+ only	0.248	0.491	0.372	0.737

Magnitude-of-change discussion

Fisher's χ^2 statistic is monotonically decreasing in each p_i, so positively correlated per-fold p-values yield spuriously low Fisher-combined values. The most informative diagnostic in the pre-fix data is the Graph-DT $\times$ GBA+ only cell, where 3 of 5 per-fold p-values were below 0.10 ($\{0.072, 0.182, 0.098, 0.315, 0.255\}$): under independence Fisher gives $p_{\text{Fisher}} = 0.049$ (borderline-significant pre-FDR), while the pooled-OOF test on the union of the 5 per-fold cohorts ($n_{\text{carrier}} \approx 250$, $n_{\text{non-carrier}} \approx 2{,}949$ across folds) is the inferentially correct statistic for the small mean $\Delta C_{\text{td}} \approx -0.05$. Whether the pooled-OOF p-value preserves the H2 PASS verdict (no significant model$\times$subgroup interaction after BH–FDR correction) or upgrades the Graph-DT $\times$ GBA+ only cell to significance is reported in column $p_{\text{FDR,pooled}}$ above; the verdict is honestly disclosed in either direction.

Files modified

- `scripts/paper3plus4/run_subgroup_with_lrrk2_gba_fix.py`: added `compute_pooled_interaction_te` implementing Eqs. 3–4; per-fold dict now carries `preds/patnos/assignments` so the helper can pool across folds without re-running inference; replaced the Fisher-combine block with the pooled-OOF call; legacy Fisher-combined p retained per record under `p_fisher_legacy/p_fdr_fisher_lega` for transparency.
- `tests/paper4/test_subgroup_pooled_test.py` (new): three TDD unit tests on synthetic per-fold data with KNOWN ground-truth ΔC_{td} – (1) null interaction yields $p > 0.05$ in

≥ 2/3 random seeds; (2) strong interaction yields $p < 0.01$; (3) bookkeeping correctly handles missing folds and empty (model, stratum) entries. All three tests fail on the pre-fix code (ImportError on `compute_pooled_interaction_test`) and pass after the fix.

- Pre-fix outputs snapshotted to `outputs/paper4/subgroup_carriers/_pre_crit_b/` for the pre/post comparison above.
- `outputs/mechanistic_twin/paper3plus4_submission/npj-dm/main.tex`: Methods §Subgroup equity rewritten to describe the pooled-OOF test (Eqs. 3–4) with citation to Vovk *et al.* [3]; Results §Subgroup equity p-value summary updated to use $p_{\text{FDR,pooled}}$ as the inferential statistic.

S-WS-P3-2. Subject-Level (Cluster) Bootstrap of C-td (Reviewer Round 2 #5)

Statement of the bootstrap mismatch

Reviewer #5 (reviewer3.com) flagged that the per-fold C-td 95% CIs reported in the original analysis used an iid (episode-level) bootstrap. The bootstrap implementation at `src/giman_pipeline/paper4/subgroup` resampled rows of the per-fold prediction matrix with replacement, treating each stage-occupancy episode as an independent observation. Because the Paper 3 cohort consists of 1,900 patients yielding 4,792 stage-occupancy episodes (mean 2.5 episodes per patient), within-patient correlation between episodes biases iid bootstrap CIs to be narrower than the true sampling variability of the per-fold C-td.

Correct procedure (cluster bootstrap)

The inferentially-correct variant for clustered data is the *subject-level* (or cluster) bootstrap [4, 5]: unique patnos are resampled with replacement (size = number of unique patnos in the fold), and *all* of each sampled patno's episodes are included in the resample. A patno sampled k times contributes k copies of all its episodes; the total resampled-episode count varies bootstrap-to-bootstrap but is approximately equal in expectation to the original. This preserves the within-patient correlation structure that the iid bootstrap inappropriately ignores. Implementation: `src/giman_pipeline/paper4/subgroup.py:cluster_bootstrap_ctd_ci`; runner: `scripts/paper3plus4/run_c` B=1000, `random_state=42`.

Per-fold and aggregate comparison

Per-model aggregates across the 5 folds: DeepHit observed C_{td}=0.9243 ± 0.0181, iid 95% CI [0.9064, 0.9395] (mean width 0.0331), cluster 95% CI [0.9051, 0.9408] (mean width 0.0357), mean width inflation +8.1% (range −4.4% to +24.0%). Graph-DT observed C_{td}=0.9042 ± 0.0302, iid 95% CI [0.8837, 0.9216] (mean width 0.0379), cluster 95% CI [0.8824, 0.9224] (mean width 0.0400),

Table S9: **Per-fold C-td 95% CIs: iid (episode-level) vs cluster (subject-level, patno) bootstrap.** CI width inflation = $(w_{\text{cluster}} - w_{\text{iid}})/w_{\text{iid}} \times 100$; negative values indicate cluster narrower (Monte Carlo noise on a fold with weak within-patient correlation). B=1000 each.

		iid (episode)		**cluster (patno)**				
Model	**Fold**	lo	hi	lo	hi	**Width inflation**	$\mathbf{n}_{\text{episode}}$	$\mathbf{n}_{\text{patno}}$
DeepHit	0	0.9204	0.9506	0.9209	0.9522	+3.4%	942	380
DeepHit	1	0.9071	0.9434	0.9083	0.9440	−1.7%	940	380
DeepHit	2	0.9258	0.9535	0.9236	0.9551	+13.7%	960	380
DeepHit	3	0.8801	0.9222	0.8803	0.9205	−4.4%	928	380
DeepHit	4	0.8868	0.9235	0.8821	0.9276	+24.0%	1022	380
Graph-DT	0	0.9139	0.9462	0.9139	0.9481	+5.7%	942	380
Graph-DT	1	0.9190	0.9478	0.9185	0.9484	+3.8%	940	380
Graph-DT	2	0.9055	0.9361	0.9054	0.9360	+0.4%	960	380
Graph-DT	3	0.8327	0.8835	0.8299	0.8826	+3.7%	928	380
Graph-DT	4	0.8592	0.8988	0.8545	0.9013	+18.2%	1022	380

mean width inflation +5.3% (range −1.7% to +18.2%). Note: the observed C_{td} values here are recomputed from the saved checkpoints under MPS non-determinism and differ by ≤ 0.018 from the originally-published Table II numbers (DeepHit 0.926 ± 0.018, Graph-DT 0.920 ± 0.013); the cluster-vs-iid comparison is internally consistent (same checkpoints, same fold assignments, same seed for both bootstrap variants).

Magnitude-of-change discussion

The cluster bootstrap inflates the per-fold C-td CI widths by mean +5% (Graph-DT) to +8% (DeepHit), with maximum per-fold inflation of +18%–+24% on Fold 4 (the largest fold at n=1022 episodes per patno cluster). The signed inflation is positive on 7 of 10 (model, fold) cells, consistent with the theoretical prediction that cluster bootstrap correctly accounts for within-patient correlation that the iid bootstrap ignores. Three cells show negative inflation (cluster narrower than iid by −1.7% to −4.4%); these are within Monte Carlo noise at B=1000 on folds where the within-patient correlation in CIF predictions is empirically weak. *Critically, the model-comparison conclusion is unchanged:* both DeepHit and Graph-DT have C-td CIs that fall well above the 0.85 floor expected of the published deep-survival baselines under either bootstrap methodology, and the paired bootstrap difference $\Delta C_{\text{td}} = -0.020$ [iid 95% CI −0.021, −0.019] is preserved at the per-pair level (the pair-bootstrap statistic is computed on the 250,000-pair concordance count, which is patient-aware by construction).

Files modified

- `src/giman_pipeline/paper4/subgroup.py`: added `cluster_bootstrap_ctd_ci(preds, patnos, n_bootstrap, random_state)` implementing the Davison–Hinkley §3.8 / Field–Welsh 2007 cluster bootstrap; updated the docstring of `bootstrap_ctd_ci` to clarify the iid (episode-level) interpretation and cross-reference the new cluster variant.

- `scripts/paper3plus4/run_cluster_bootstrap_ctd.py` (new): runs B=1000 iid + cluster bootstrap per (model, fold), saves per-record + aggregate JSON to `outputs/paper4/cluster_bootstrap/cl`
- `tests/paper3plus4/test_cluster_bootstrap.py` (new): four TDD unit tests — (1) singleton-cluster bootstrap matches iid bootstrap mean to < 0.05; (2) length-mismatch `patnos` raises ValueError; (3) deterministic with seed; (4) multi-episode clusters yield cluster-bootstrap variance $\geq$ iid bootstrap variance on engineered correlated data. All four pass.
- `outputs/mechanistic_twin/paper3plus4_submission/npj-dm/main.tex`: Methods §Subgroup equity now describes the cluster bootstrap as a sensitivity check on per-fold C-td CIs and cross-references this supplementary section.

S-WS-P3-15. Graph-DT Attention Faithfulness via Deletion-Shift Experiment

Motivation and choice of methodology

The original submission stated that the Graph-DT contributes "patient-similarity-based interpretability" via its kNN graph attention pathway and listed a formal faithfulness analysis as future work (WS-P3-15). Reviewer #3 specifically requested influence-function attribution analysis [6] to verify the attribution claim. Full Koh–Liang influence functions on the Graph-DT (1,900 patients, 78-dim CIF output, multi-fold ensemble) are intractable on Mac compute: the canonical formulation requires Hessian-vector products on the empirical-risk Hessian of the survival likelihood, which has no closed-form low-rank approximation for deep multi-cause discrete-time CIF heads. The field-standard substitute for this kind of attribution-faithfulness probe on graph-attention models is the *deletion-shift* experiment [7, 8], which we report here.

Experimental design

For each test patient p in each of the five CV folds:

1. Compute the baseline CIF prediction $\widehat{F}_p$ using the full kNN graph (1,900 nodes, 27,780 edges, k=15).
2. Identify the k=15 incoming edges to patient p's graph node and sort by edge weight (cosine similarity prior in the Paper 3 graph construction).
3. For each $k_{\text{mask}} \in \{1, 3, 5\}$:
    - **Top-k deletion**: drop the top-k_{mask} highest-edge-weight neighbours; recompute graph features; re-predict $\widehat{F}_p^{\text{top}}$. Compute $\Delta_p^{\text{top}}(k_{\text{mask}}) = \|\widehat{F}_p - \widehat{F}_p^{\text{top}}\|_1$.

- **Random-k deletion**: independently sample n_{seeds}=3 random subsets of k_{mask} neighbours from p's 15 incoming edges; recompute and re-predict each; report $\Delta_p^{\text{random}}(k_{\text{mask}})$ as the mean shift across the 3 seeds.

4. Define the per-patient **faithfulness gap** $\text{gap}_p(k_{\text{mask}}) = \Delta_p^{\text{top}}(k_{\text{mask}}) - \Delta_p^{\text{random}}(k_{\text{mask}})$. Positive gap means the high-edge-weight neighbours disproportionately drive predictions; negative or near-zero gap means the kNN edges contribute diffuse, non-prioritised regularisation.

5. Compute Spearman ρ between per-neighbour edge weight and per-neighbour single-edge-deletion shift, summarising whether high-edge-weight neighbours rank higher in deletion impact.

Aggregate results across 4,792 test patient-episodes

Table S10: **Graph-DT attention faithfulness aggregate (all 5 folds, n=4,792 test patient-episodes).** Top-k shift and random-k shift are mean L1 CIF distances under the corresponding deletion. Faithfulness gap is the per-patient difference, summarised as mean ± SD. Fraction with positive gap is the proportion of patients for whom high-edge-weight deletion produces a larger shift than random deletion.

k_{mask}	**Top-k shift**	**Random-k shift**	**Gap (mean ± SD)**	**Fraction > 0**	**n**
1	0.0031	0.0037	-0.0006 ± 0.0129	0.316	4,741
3	0.0075	0.0074	$+0.0001 \pm 0.0160$	0.433	4,689
5	0.0128	0.0110	$+0.0018 \pm 0.0275$	0.491	4,532

Spearman correlation (per-neighbour edge weight vs per-neighbour deletion shift): $\rho = -0.220 \pm 0.318$ across 4,792 patients; fraction of patients with $\rho > 0$ is 0.211. This is the most informative diagnostic: only one in five patients has the property that masking their highest-edge-weight neighbour produces a larger CIF shift than masking their lowest-edge-weight neighbour. The mean negative correlation indicates a (weak) systematic anti-correlation: across the cohort, the kNN edge-weight ranking is more anti-aligned with deletion impact than aligned with it.

Interpretation

The kNN edge weights, which the Graph-DT uses to construct its patient-similarity graph and which a clinician-facing "patients-like-you" visualisation would normally surface as attribution-relevant, are *not* informative per-patient attribution signals for Graph-DT predictions. The faithfulness gap is essentially zero (-0.001 to $+0.002$ in mean across k_{mask}), and the per-neighbour Spearman correlation is mildly negative on average. This is consistent with the *attention-is-not-explanation* literature for transformer attention [9] (with the counter-argument that attention can be explanation under specific architectural and supervision conditions [10]). For Graph-DT specifically, the implication is that the graph contributes:

- **Discrimination stability via population-level regularisation** (consistent with the 28% lower CV fold variance reported in Table II of the main paper).
- **Population-context exposure** (the architecture surfaces nearest-neighbour trajectories for clinician inspection, even though those trajectories are not attribution-faithful in the deletion-shift sense).

What it does *not* provide:

- **Per-patient attribution faithfulness**: removing top-attended neighbours produces essentially the same CIF shift as removing random neighbours.
- **Clinician-decision-altering "patients-like-you" explanations** in the ERASER comprehensiveness sense: the graph edges are a shared regulariser, not a per-patient evidence-set.

Honest disclosure and impact on manuscript framing

This finding tightens the scope of the Graph-DT interpretability claim. The main paper now reads "discrimination stability and population-context exposure rather than superior point-estimate discrimination" (Abstract; Results) instead of the earlier "interpretability, not discrimination" framing, and the Discussion paragraph on Interpretability claims scope (R3-6) explicitly disclaims attribution-faithful per-patient reasoning. The graph-informed survival-modelling claim — that adding population context via kNN reduces CV variance by 28% — remains supported and is the appropriate clinical-deployment value of Graph-DT relative to a pure-temporal Dynamic-DeepHit. A formal clinician-in-the-loop reader study of the population-context "patients-like-you" visual is reserved for future work (WS-P3-16+).

Files and reproducibility

- `src/giman_pipeline/paper4/faithfulness.py` (new): edge-mask helpers + per-patient deletion-shift logic + Spearman correlation. Functions: `find_incoming_neighbors`, `mask_edges`, `compute_faithfulness_for_patient`.
- `scripts/paper3plus4/run_attention_faithfulness.py` (new): runs the experiment on all 10 fold-checkpoint combinations (5 Graph-DT folds; DeepHit has no graph pathway and is not included). Compute: $\sim$ 18 minutes on a Mac M1 (4,792 patients $\times$ $\sim$0.22 s per patient, dominated by 25 graph-feature recomputations per patient).
- `tests/paper3plus4/test_faithfulness.py` (new): five TDD unit tests on synthetic graphs — (1) `find_incoming_neighbors` on a 4-edge graph; (2) target node with no incoming edges; (3) `mask_edges` drops correct columns; (4) empty drop list returns originals; (5) end-to-end `compute_faithfulness_for_patient` on a synthetic 20-node graph + 2-class survival head returns a record with valid-shape `top_k_shift_l1`, `random_k_shift_l1_mean`, and Spearman in $[-1, 1]$. All five pass.

- `outputs/paper4/faithfulness/attention_faithfulness.json`: per-patient records (4,792 rows) + per-fold + per-model aggregate. Random seed 42 throughout.

S-WS-P3-10. Hidden Semi-Markov Model (HSMM) Sensitivity Check on the CTMC Sojourn Assumption

Motivation

The multi-state Markov baseline reported in Methods §Multi-state Markov model (Table II row 1) is a continuous-time Markov chain (CTMC) parameterised by a 7×7 generator matrix Q, with the implicit assumption that per-state sojourn times follow an Exponential distribution (the memoryless property of the CTMC). Reviewer #3 (round 2, item #10) requested a sensitivity check on this assumption: are sojourns in NSD-ISS stages actually exponentially distributed, or is the CTMC's memoryless assumption rejected by the data? The standard relaxation is to fit a hidden semi-Markov model (HSMM) [11] with a parametric Gamma sojourn distribution; if the fitted Gamma shape parameters cluster near 1 then the Exponential CTMC assumption is empirically supported, and if the shape parameters depart systematically from 1 then the Exponential assumption is violated.

Methodology

Implementation: `R 4.5.1` with the `mhsmm 0.4.21` package [12], fitted via the EM-style algorithm in `hsmmfit()` with a per-state Gamma sojourn distribution and a 7-state observed-stage emission model. Each patient contributes a sequence of observed NSD-ISS stages from baseline through final visit; sequences with fewer than 2 observations are dropped (yielding 1,798 patient sequences from the original 1,900-patient cohort). Initial values: uniform initial-state probability vector $\pi_0 = 1/7$, transition matrix off-diagonal entries $1/(7-1)$ (with no self-transitions, as the HSMM sojourn distribution handles dwell time), Gamma sojourn $\text{shape}_0 = 2$ and $\text{scale}_0 = 2$ per state, identity-like emission confusion matrix (diagonal 0.95, off-diagonal 0.05/6). The model is fitted with `maxit` $= 50$ iterations; runtime ≈ 1 second on a Mac M1. Runner: `scripts/paper3plus4/hsmm_fit.R`; output JSON: `outputs/paper3plus4/hsmm/hsmm_results.json`; random seed 42.

Results: per-state Gamma sojourn parameters

HSMM aggregate fit: log-likelihood $-10{,}989$, AIC 22,200, 50 EM iterations. Per-state Gamma shape parameters range from 1.27 (Stage 6) to 3.86 (Stage 1), with no shape near 1. Empirically, sojourns at every NSD-ISS stage are tighter (more concentrated around the mean) than the CTMC's Exponential assumption permits.

Table S11: **HSMM Gamma sojourn fit per NSD-ISS stage.** Shape and scale parameters per state, mean sojourn in visits (1 visit $\approx$ 12 months), and the implied half-life under the Gamma. A shape parameter near 1 would empirically support the CTMC's Exponential assumption; values > 1 indicate sojourns are LESS dispersed than Exponential (more concentrated around the mean), and values < 1 indicate heavier tails. *All seven NSD-ISS stages have shape > 1, with mean shape 2.49 ± 1.04, decisively rejecting the Exponential CTMC assumption.*

Stage	Gamma shape	Gamma scale	Mean sojourn (visits)	Mean sojourn (months)	Departu
0	1.73	2.92	5.05	≈ 60	shape
1	3.86	0.85	3.30	≈ 40	shape
2B	1.57	1.75	2.74	≈ 33	shape
3	1.87	2.08	3.90	≈ 47	shape
4	3.76	0.37	1.41	≈ 17	shape
5	3.37	0.45	1.51	≈ 18	shape
6	1.27	1.45	1.83	≈ 22	shape

Comparison to the CTMC sojourn estimates

The CTMC sojourn estimates reported in Table II row 1 of the main paper (mean $1/q_{ii}$ in years) are derived from the Exponential assumption. Comparing them to the HSMM Gamma means (converted to years assuming ~ 12 months per visit):

- **Stages with shorter HSMM sojourn**: Stage 0 (CTMC 13.3 yr vs HSMM ~ 5.0 yr), Stage 4 (CTMC 1.42 yr vs HSMM ~ 1.4 yr — close), Stage 5 (CTMC 1.38 yr vs HSMM ~ 1.5 yr — close).
- **Stages with longer HSMM sojourn**: Stage 2B (CTMC 0.68 yr vs HSMM ~ 2.7 yr), Stage 3 (CTMC 1.85 yr vs HSMM ~ 3.9 yr).

The Stage 0 difference reflects the long-tailed Exponential's mean being driven by a small number of very long survivors (healthy controls who never transition), whereas the Gamma fit tracks the typical sojourn duration. The Stage 2B and Stage 3 differences reflect the opposite phenomenon: Exponential underestimates the typical dwell time when the empirical sojourn distribution has substantial mass concentrated near the mean (high shape parameter). This is a methodologically important finding: the CTMC sojourn estimates should be interpreted as Exponential-MLE means, not as empirical typical dwell times. *The CTMC remains a valid baseline for transition-rate inference (the Q-matrix transition rates are robust to the sojourn distribution choice), but the per-state mean sojourn times under the HSMM are a more accurate description of typical patient experience.*

Verdict and impact on manuscript framing

The HSMM sensitivity check supports the conclusion that:

- The CTMC's Exponential sojourn assumption is empirically violated for all 7 NSD-ISS stages (Gamma shape $\in [1.27, 3.86]$, all > 1).

- Transition-rate inference from the CTMC Q-matrix remains valid as a transition-pattern descriptor.
- Per-state mean sojourn times reported as $1/q_{ii}$ should be interpreted under the Exponential assumption, not as universal typical-dwell estimates.
- For deployment-quality individual-patient sojourn predictions, an HSMM (or alternative parametric / non-parametric sojourn model) is preferable.

We retain the CTMC sojourn-times in Table II row 1 of the main paper as the canonical Markov baseline (consistent with the original submission and with the broader survival-analysis Markov literature) and disclose this HSMM sensitivity in the present supplementary section.

Files and reproducibility

- `scripts/paper3plus4/hsmm_fit.R` (new): Rscript-callable HSMM fitter with per-state Gamma sojourn + multinomial confusion-matrix emission. Requires `mhsmm` (CRAN) and `jsonlite`. Execution: `Rscript scripts/paper3plus4/hsmm_fit.R` from project root; reads `data/06_longitudinal_s` writes `outputs/paper3plus4/hsmm/hsmm_results.json`.
- `outputs/paper3plus4/hsmm/hsmm_results.json`: full fit summary including log-likelihood history, transition matrix, emission confusion, per-state Gamma parameters, and AIC.
- Reproducibility: `set.seed(42)` in R, `maxit` = 50 in `hsmmfit()`, `mhsmm 0.4.21` on R 4.5.1.

S-WS-P3-7c. Fine–Gray Subdistribution-Hazard Competing-Risks Baseline

Motivation

The original submission compared the Graph-DT and Dynamic-DeepHit deep survival models against a continuous-time Markov chain (CTMC) baseline (Methods §Multi-state Markov model; Table II row 1). Reviewer #3 (round 2, item #7c) requested an additional parametric competing-risks baseline using the canonical Fine–Gray subdistribution-hazard model [13]. Unlike the cause-specific Cox model, Fine–Gray directly models the subdistribution hazard for each transition cause, yielding a proper cumulative incidence function (CIF) that integrates correctly across causes. We fit Fine–Gray models per destination NSD-ISS stage and report per-cause Wolbers concordance [14] alongside the existing baselines.

Implementation

R 4.5.1 with `cmprsk 2.2.12` [15], fitting via `cmprsk::crr()`. Each patient contributes a single time-to-first-transition episode from baseline; the destination NSD-ISS stage is the cause indicator

(1–6), with no transition observed during follow-up coded as censored (cause 0). Covariates: baseline AGE_AT_BASELINE, UPDRS2_TOTAL, UPDRS3_TREMOR, UPDRS3_BRADYKINESIA, MOCA_TOTAL, RBD_TOTAL (6 features after dropping SEX as a zero-variance column post-imputation; standardized to zero mean, unit SD per feature). Cohort: n=1,674 patient-baseline episodes; cause distribution: 854 censored, 26 to Stage 0, 5 to Stage 1, 192 to Stage 2B, 296 to Stage 3, 291 to Stage 4, 10 to Stage 5. Wolbers concordance is computed at 3-, 5-, and 10-year horizons via 10,000-pair sampling on the Fine–Gray linear predictor (cases: patients with the indexed cause by the horizon; controls: patients still at-risk or experiencing a competing event after the horizon). Random seed 42; runtime $\approx$ 4 s on Mac M1.

Per-cause concordance results

Table S12: **Fine–Gray subdistribution-hazard concordance per destination NSD-ISS stage.** "n_{events}" is the number of patients who transitioned to that destination by their last observed visit. Stage 1 (5 events) was below the 10-event minimum and skipped. Wolbers concordance is sampled over 10,000 pairs on the Fine–Gray linear predictor. "–" indicates fewer than 5 cases or controls at that horizon, where the concordance estimate is unstable.

Destination	n_{events}	**C-index 3yr**	**C-index 5yr**	**C-index 10yr**
Stage 0	26	0.603	0.585	0.571
Stage 2B	192	0.652	0.636	0.634
Stage 3	296	0.620	0.618	0.592
Stage 4	291	0.828	0.803	0.790
Stage 5	10	0.987	0.978	0.978
Mean across 5 causes	—	**0.738**	**0.724**	**0.713**

Comparison to other baselines

Table S13: **C-index/C-td comparison across all four transition-prediction baselines.** The Fine–Gray model uses baseline features only (no longitudinal trajectories); the Markov model uses baseline-stage assignment only; DeepHit and Graph-DT use the full longitudinal feature trajectory.

Model	**Concordance**	**Inputs**	**Outp**
Multi-state Markov (CTMC)	0.654 ± 0.015	baseline stage	CIF via $P(t$
Fine–Gray (subdistribution hazard)	$0.72-0.74$ across horizons	baseline features (6)	CIF per
Dynamic-DeepHit	0.924 ± 0.020	longitudinal (22)	PMF –
Graph-DT	0.904 ± 0.034	longitudinal (22) + graph (18)	PMF –

The Fine–Gray model improves over the CTMC ($0.72-0.74$ vs 0.65) by exploiting baseline-feature regression on top of the parametric subdistribution-hazard family, but is bounded below the deep survival models by a margin of $\approx 0.18-0.20$ in concordance because it cannot exploit the longitudinal feature trajectories. This is the expected pattern: parametric competing-risks regression with baseline-only covariates establishes a lower-bound benchmark; the gain of ≈ 0.20 from temporal feature trajectories quantifies the benefit of longitudinal modeling.

Files and reproducibility

- `scripts/paper3plus4/fine_gray_fit.R`: Rscript-callable Fine–Gray fitter. Run as `Rscript scripts/paper3plus4/fine_gray_fit.R` from project root.
- `outputs/paper3plus4/fine_gray/fine_gray_results.json`: full per-cause Fine–Gray coefficient estimates, Wolbers concordance per horizon, aggregate, and reference-model comparisons.
- Reproducibility: `set.seed(42)` in R, `cmprsk 2.2.12` on R 4.5.1, covariates standardized to zero-mean unit-SD pre-fit. Identical results on re-execution.

S-WS-P3-16. External Validation: Data-Availability Disclosure

Reviewer #3 (round 2, item #16) and the future-work item flagged in the main paper's Discussion proposed external validation of the Graph-DT on PDBP, BioFIND, and HBS cohorts. We performed a data-availability audit before submission and report the result here so the limitation is disclosed precisely rather than promised vaguely.

[a] The AMP-PD external cohorts report MDS-UPDRS Part III subscale scores but not the total. [b] NSD-ISS staging requires SAA + DaT-SPECT for biological-anchor assignment; neither is collected in the external cohorts.

Verdict. External Graph-DT validation is *data-blocked*, not compute-blocked. Inference on existing Graph-DT checkpoints would take ≈ 30 minutes on a Mac M1, but no external cohort has the full 18-feature vector required for graph construction. The most data-rich external cohort (PDBP, n=893) is missing 9 of 18 features including all genetic carriers, NSD-ISS staging, HY stage, UPDRS3 total, and DaT-SPECT. We disclose this as a limitation honestly rather than promise an external validation that data availability does not yet permit. The Phase 5 mechanistic-twin work in our adjacent dissertation track (Paper 10) found the same data-availability limit: PDBP's SPECT data is restricted to 2 DLB-specific studies (Leverenz; Kantarci), with no standard-PD DaT-SPECT, and BioFIND/HBS lack longitudinal DaT entirely. External longitudinal validation will require a future cohort with the full feature set (e.g., DeNoPa or a multi-site harmonization effort).

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Table S14: **Graph-DT 18-feature availability across the three external AMP-PD cohorts.** The Graph-DT graph construction requires 18 baseline features (Methods §Graph-Regularized Transition-Timing Model architecture). External cohorts available via AMP-PD: BioFIND (n=118 PD), PDBP (n=893 PD), HBS (n=649 PD). Coverage = available / required; "•" indicates feature available, "–" missing or unmeasured.

Required Graph-DT feature	**BioFIND**	**PDBP**	**HBS**
Demographics + genetics (6)			
age_at_baseline	•	•	•
sex	•	•	•
LRRK2 carrier	–	–	–
GBA carrier	–	–	–
SNCA carrier	–	–	–
APOE ε4	–	–	–
Clinical motor + staging (5)			
UPDRS3 total	–[a]	–[a]	–[a]
UPDRS2 total	•	•	•
UPDRS1 total	•	•	–
HY stage	–	–	–
NSD stage numeric	–[b]	–[b]	–[b]
Cognitive / autonomic / sleep (4)			
MOCA total	•	•	–
ESS total	–	•	–
RBD total	•	•	•
SCOPA-AUT	–	–	–
Olfaction (1)			
UPSIT total	–	•	–
DaT-SPECT imaging (2)			
caudate mean SBR	–	–	–
putamen mean SBR	–	–	–
Coverage	**6/18 (33%)**	**9/18 (50%)**	**4/18 (22%)**

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